

Def. Let $A \in M_{m \times n}(F)$ be a m by n matrix

The first elementary Row operation is: for any transposition $(i \ j) \in S_m$,

$$E_{i,j}^{1st} \stackrel{\text{def}}{=} \begin{bmatrix} e_{(i,j)(1)} \\ e_{(i,j)(2)} \\ \vdots \\ e_{(i,j)(m)} \end{bmatrix} \quad (E_{i,j}^{1st} A \text{ is a } i\text{th and } j\text{th row-changed matrix.})$$

The second elementary Row operation is: for any non-zero scalar $c \in F$, and $k \in \{1, \dots, m\}$

$$E_c^{2nd} \stackrel{\text{def}}{=} \begin{bmatrix} e_1 \\ \vdots \\ c \cdot e_k \\ \vdots \\ e_m \end{bmatrix} \quad (E_c^{2nd} \cdot A \text{ is multiplying } c \text{ to } k\text{th row})$$

The third elementary Row operation is: for any scalar $c \in F$ and $1 \leq i \neq j \leq m$,

$$E_{c,i,j}^{3rd} \stackrel{\text{def}}{=} \begin{bmatrix} e_1 \\ \vdots \\ c \cdot e_i + e_j \\ \vdots \\ e_m \end{bmatrix} \leftarrow j\text{th column.} \quad (E_{c,i,j}^{3rd} A \text{ is } c \cdot (i\text{th row}) \text{ adding to } j\text{th row.})$$

Moreover, these are all invertible,

$$(E_{i,j}^{1st})^{-1} = E_{i,j}^{1st}, \quad (E_c^{2nd})^{-1} = E_{c^{-1}}, \quad (E_{c,i,j}^{3rd})^{-1} = E_{-c,i,j}^{3rd}$$

$$\begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ c \cdot e_i + e_j \\ \vdots \\ e_m \end{pmatrix} \begin{pmatrix} A_1 \\ \vdots \\ A_n \end{pmatrix} = \begin{pmatrix} A_1 \\ \vdots \\ A_j + c \cdot A_i \\ \vdots \\ A_n \end{pmatrix}$$

$$\vdots \vdots \vdots A_i =$$

$$A_j + c \cdot A_i$$

Rank of a Matrix.

Def. Given a Matrix $A \in M_{m \times n}(F)$, define

$$\text{rank } A \stackrel{\text{def}}{=} \dim \text{Im } L_A.$$

Clearly, $\text{rank } A \leq m$.

Prop. Let $A \in M_{m \times n}$, and $P \in M_{m \times m}$ and $Q \in M_{n \times n}$ be invertibles, Then

$$\text{rank } PAQ = \text{rank } A.$$

Proof. Observe that

$$\text{Im } L_{PAQ} = \text{Im}(L_P \circ L_A \circ L_Q) = (L_P \circ L_A \circ L_Q)[F^n] = (L_P \circ L_A) \underbrace{[F^n]}_{L_Q \text{ onto}} = [F^n]$$

Moreover, since L_P is an Isomorphism, $L_P[L_A[F^n]] \cong L_A[F^n] = \text{Im } L_A$. $\square$

Thm. Given a Matrix $A \in M_{m \times n}$,

$$\text{rank } A = \dim \text{span}\{a_1, \dots, a_n\} \text{ where } a_i \text{ is the } i\text{th column of } A. \\ (\text{or } a_i = Ae_i)$$

Proof. $\text{rank } A = \text{rank } L_A = \dim \text{Im } L_A = \dim \text{span}\{L_A(e_1), \dots, L_A(e_n)\} = \dim \text{span}\{a_1, \dots, a_n\}$.

Thm. Given a Matrix $A \in M_{m \times n}$ of rank $A = r$, then there are invertibles $B \in M_{m \times m}$, $C \in M_{n \times n}$,

$$\text{such that } BAC = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

Thm. Let $T: V \rightarrow W$ and $U: W \rightarrow Z$ be linear maps on fin. dim V, W, Z . Then

$$\text{rank } U \circ T \leq \text{rank } U \text{ and } \text{rank } U \circ T \leq \text{rank } T.$$

Proof. $\text{Im } U \circ T = (U \circ T)[V] \subseteq U[W] = \text{Im } U$, so $\text{rank}(U \circ T) \leq \text{rank } U$.

$$\text{So, } \text{rank}(U \circ T) = \text{rank}(T^t \circ U^t) \leq \text{rank } T^t = \text{rank } T \quad \square$$

Def. Given $A \in M_{m \times n}(F)$, Consider the linear map

$$L_A: F^n \rightarrow F^m; X \mapsto AX$$

Then, $\ker L_A$ is the solution set of the homogeneous linear system $AX=0$.

Thm. For any kind elementary Row operation E ,

$$\ker L_A = \ker L_{EA}.$$

Proof. $AX=0$, then clearly $EAX = E0 = 0$. So $\ker L_A \subseteq \ker L_{EA}$.

Conversely, since E is invertible, $EAX=0 \Rightarrow AX=E^{-1}0=0$.

Thm. Given a m by n matrix $A \in M_{m \times n}(F)$ and $B \in F^m$,

if $S \in F^n$ satisfies $AS=B$, then

$$x \in F^n \text{ satisfies } Ax=B \Leftrightarrow x \in S + \ker L_A.$$

Proof. If $Ax=B$, then $Ax=B=AS$, so $A(x-s)=0$. Thus $x-s \in \ker L_A$, or $x \in S + \ker L_A$.

Conversely, $x \in S + \ker L_A$ be given. That is, $x = s + \gamma$ for some $\gamma \in \ker L_A$, so

$$Ax = A(s + \gamma) = B + 0 = B.$$

Corollary. $Ax=b$ has only one solution iff A is invertible.

Corollary. Given linear system $Ax=b$ where $A \in M_{m \times n}$, $x \in F^n$, $b \in F^m$,

$$Ax=b \text{ has a solution if and only if } \operatorname{rank} A = \operatorname{rank}(A|b).$$

Proof. Since $Ax=b$ for some $x \in F^n$ if and only if $b \in \operatorname{Im} A$, and

$$\operatorname{Im} A = \operatorname{span}\{Ae_1, \dots, Ae_n\}$$

So, proved. $\square$

In Inner Product space, we can use inner product to find "best" approximation solution.

Thm. Let $A \in M_{m \times n}(\mathbb{C})$ and $y \in \mathbb{C}^m$ be given.

Then, there exists $x_0 \in \mathbb{C}^n$ such that

$$A^* A x_0 = A^* y \quad \text{and} \quad \|A x_0 - y\| \leq \|A x - y\| \text{ for all } x \in \mathbb{C}^n.$$

In particular, if $\text{rank } A = n$, then $x_0 = (A^* A)^{-1} A^* y$.

Proof. Given $y \in \mathbb{C}^m$, there exists an orthogonal projection $A x_0$ on $\text{Im } A$.

That is, $\langle A x_0 - y, A x \rangle = 0$ for all $x \in \mathbb{C}^n$.

And, since $\langle A x_0 - y, A x \rangle = \langle A^* A x_0 - A^* y, x \rangle$, so $A^* A x_0 - A^* y = 0$.

(Specifically, for orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$ for $\text{Im } A \subseteq \mathbb{C}^m$,

$$A x_0 = \sum_{i=1}^n \langle y, \alpha_i \rangle \cdot \alpha_i.)$$

Moreover, if $\text{rank } A = n$, then $\text{rank } A^* A = n$, so invertible.

System of linear equations: perspective from linear functional.
 Let $V = F^n$ and $W \leq V$ be a subspace. There exists a basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_k\}$ for W ,
 and let $\overline{\mathcal{B}} = \mathcal{B} \cup \{\alpha_{k+1}, \dots, \alpha_n\}$ be an extended basis for V .
 For each $1 \leq j \leq n$, define a linear functional on V :

$$f_j: V \rightarrow F: \sum_{i=1}^n c_i \alpha_i \mapsto c_j.$$

(That is, $\{f_1, \dots, f_n\}$ is a dual basis for V^* corresponding to $\overline{\mathcal{B}}$.)

Then, $W = \text{Span}\{\alpha_1, \dots, \alpha_k\} = \ker f_{k+1} \cap \dots \cap \ker f_n$.

Now, let $A = \begin{bmatrix} [f_{k+1}]_S^T \\ [f_{k+2}]_S^T \\ \vdots \\ [f_n]_S^T \end{bmatrix}$ where $S = \{\alpha_1, \dots, \alpha_n\}$ is a standard basis for V .
 $[f_n]_S^T \in M_{(n-k) \times n}$
 satisfies that $\ker A = W$.

Problem. Find a matrix A such that